FOODORA - A FOOD BILLING APPLICATION

Submitted by

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BONAFIDE CERTIFICATE

Certified that this Project titled "Foodora - A Food Billing Application" is the Bonafide work of "YOKESHWARAN K (211623070189) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

Submitted for the End semester practical examination to be held on __

INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

Foodora is a mobile-based food ordering and billing application developed to modernize food service management in college canteens and cafeterias. Traditional canteen systems often rely on manual billing and order processing methods, which lead to long queues, billing errors, inefficient inventory management, and delayed service. The Foodora application addresses these challenges by providing a complete digital platform for food ordering, payment processing, and QR-code-based billing.

The application is developed using Flutter and Dart for frontend mobile development, while Firebase services such as Firestore Database and Firebase Authentication are used for backend operations and real-time data synchronization. Razorpay and UPI integration provide secure online payment facilities for users. The system enables students and staff to browse food menus, search food items, add products to cart, place orders digitally, and receive QR-code-based bills for food collection and verification.

Foodora also includes an admin management module that allows canteen administrators to manage food items, update stock quantities, monitor live orders, and maintain inventory records efficiently. The system automatically updates food stock after every successful transaction, reducing manual work and improving inventory accuracy.

The project aims to improve operational efficiency, reduce waiting time, enhance customer convenience, and provide a user-friendly food ordering experience within educational institutions. The implementation of real-time synchronization, secure payment processing, and automated inventory management makes Foodora a scalable and efficient solution for modern campus food service systems.

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CHAPTER 1: INTRODUCTION

The increasing adoption of mobile technology in everyday life has opened new possibilities for digitizing traditional processes. One such area is food ordering and management in college canteens, which has historically relied on manual processes prone to delays, billing errors, and inventory mismatches.

Foodora is a mobile application developed to address these challenges by providing a seamless, digital food ordering and billing platform specifically designed for college campus canteens. The application enables students and staff to browse menus, place orders, make secure payments, and receive QR-code-based bills — all from their smartphones.

This report presents the design, development, implementation, and testing of the Foodora system. It covers the problem statement, objectives, system architecture, module descriptions, experimental results, and future enhancement plans.

1.1 Motivation

College canteens serve hundreds of students and staff daily. Traditional ordering methods create long queues, slow service, and frequent billing mistakes. There is a clear need for a fast, reliable, and contactless food ordering solution that also helps canteen administrators manage inventory and track orders in real time.

1.2 Scope of the Project

The scope of Foodora includes:

- Mobile-based food ordering for students and staff
- Real-time stock management for canteen administrators
- Secure digital payment through Razorpay and UPI
- QR-code-based contactless billing
- Order history management and tracking

CHAPTER 2: LITERATURE SURVEY

A review of existing literature reveals several approaches to digital food ordering, restaurant management, and mobile payment systems. The following summarizes key related works.

Liyanage et al. [1] proposed Foody, a smart restaurant management and ordering system that integrates real-time inventory tracking and mobile ordering. Their work demonstrated significant reductions in service time but lacked QR-code billing integration.

Hidayat et al. [2] developed a menu ordering application using QR codes with user experience analysis. The study confirmed user satisfaction with contactless ordering but was limited to static menu display without dynamic stock updates.

Coching et al. [3] explored ingredient inventory and revenue optimization using Integer Linear Programming for restaurant food sales. Their model improved stock management but required significant computational resources unsuitable for mobile platforms.

Singhal and Konguvel [4] presented a restaurant management system designed for the COVID-19 context, emphasizing contactless interactions and hygiene protocols. Their system influenced the contactless ordering approach adopted in Foodora.

Gawaran et al. [5] focused on real-time innovation in quick-service restaurant experiences, highlighting the importance of fast order processing and live status updates — features central to Foodora's design.

Sam et al. [6] implemented a mobile application ordering system to optimize user experience, identifying intuitive UI design and minimal loading times as critical success factors.

Deshmukh et al. [7] examined streamlining food takeaway at food courts using collaborative filtering and analytics, providing insights into user preference modelling applicable to future Foodora enhancements.

Mishra et al. [8] developed a touch-based digital ordering system on Android using GSM and Bluetooth, serving as an early reference for mobile-based canteen ordering.

Anggreainy et al. [9] proposed an online food ordering system for food courts using the Scrum method, demonstrating agile development practices applicable to the iterative development of Foodora.

2.1 Research Gap

While existing systems address individual aspects such as mobile ordering, inventory management, or QR-code billing, none integrates all these features into a single mobile platform tailored for college campus canteens. Foodora bridges this gap by combining food ordering, QR-code billing, Razorpay payment integration, and real-time Firebase synchronization in one unified application.

CHAPTER 3: EXISTING SYSTEM

3.1 Overview of Existing Systems

College canteen food ordering has traditionally been managed through two primary approaches:

3.1.1 Manual Ordering System

In the manual method, students place verbal orders at the counter, canteen staff prepare handwritten bills, and payments are made in cash. This method is highly error-prone, slow, and incapable of providing real-time inventory data.

3.1.2 Kiosk-Based Ordering System

Some institutions have adopted kiosk machines for self-service ordering. While faster than manual methods, kiosks require significant hardware investment, cannot be updated remotely in real time, and still require physical interaction — a limitation highlighted during and after the COVID-19 pandemic.

3.2 Drawbacks of Existing Systems

- Long queues and high waiting times during peak hours
- Manual billing errors leading to revenue and inventory losses
- No real-time stock tracking or automated inventory management
- Limited or no support for digital payment methods
- No order history or QR-code-based bill verification
- High maintenance costs for kiosk hardware
- Inability to remotely update menus or prices

CHAPTER 4: PROPOSED SYSTEM

The proposed Foodora system is designed to provide a complete digital solution for food ordering and billing in college canteens. The system integrates food browsing, digital payments, QR-code billing, and real-time stock management into a single mobile application. The application eliminates the limitations of traditional manual and kiosk-based ordering systems by providing a faster, more accurate, and user-friendly experience.

Foodora allows students and staff to place orders directly through their smartphones without standing in queues. The system automatically updates stock quantities whenever an order is placed, preventing over-ordering and improving inventory accuracy.

4.1 Key Features of the Proposed System

- Mobile-based food ordering
- Real-time inventory tracking
- QR-code-based billing system
- Secure online payment integration
- Automated stock deduction
- User-friendly admin management system
- Order history and tracking
- Real-time synchronization using Firebase

The system uses Firebase Realtime Database for instant data synchronization and Razorpay for secure payment processing. QR-code generation enables quick order verification and contactless food collection.

The proposed system reduces manual errors, improves operational efficiency, minimizes waiting time, and enhances the overall campus dining experience.

4.2 System Architecture

The Foodora application follows a client-server architecture built on Firebase as the backend-as-a-service platform. The mobile frontend communicates with Firebase Realtime Database for order and inventory data, and with Razorpay APIs for secure payment processing.

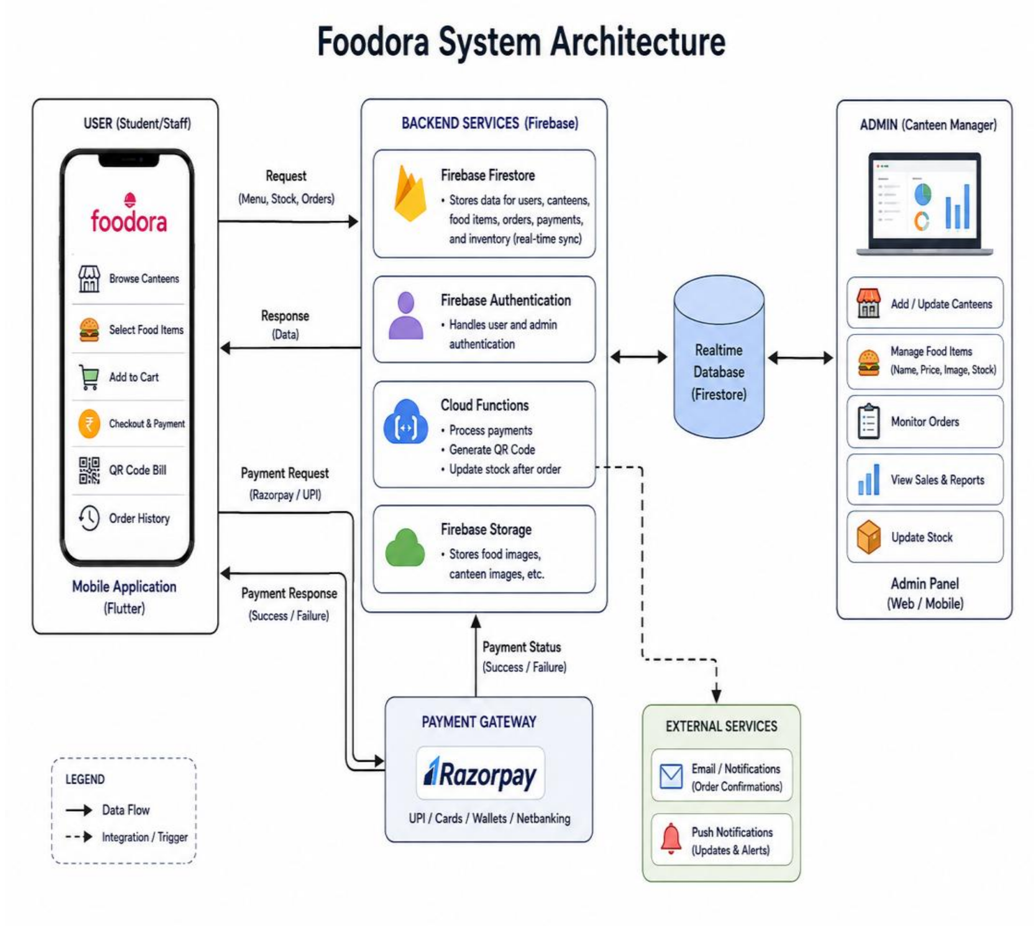


Figure 4.1 – Foodora System Architecture Diagram

4.3 Technology Stack

The following technologies are used in the development of Foodora:

Component	Technology Used
Frontend	Flutter / Dart
Backend / Database	Firebase Realtime Database
Authentication	Firebase Authentication
Payment Gateway	Razorpay SDK / UPI Integration
QR Code	ZXing / QR Code Generation Library
IDE	Android Studio
Version Control	Git / GitHub

Table 4.1 – Technology Stack

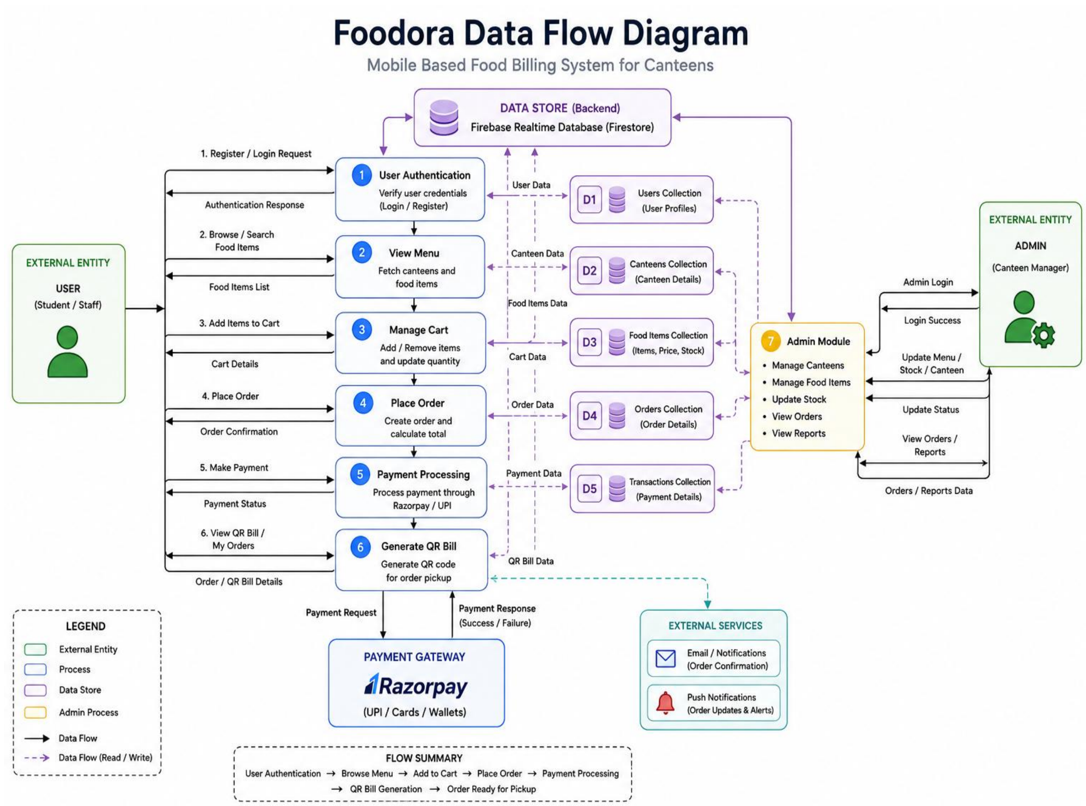


Figure 4.2 – Foodora Data Flow Diagram (DFD)

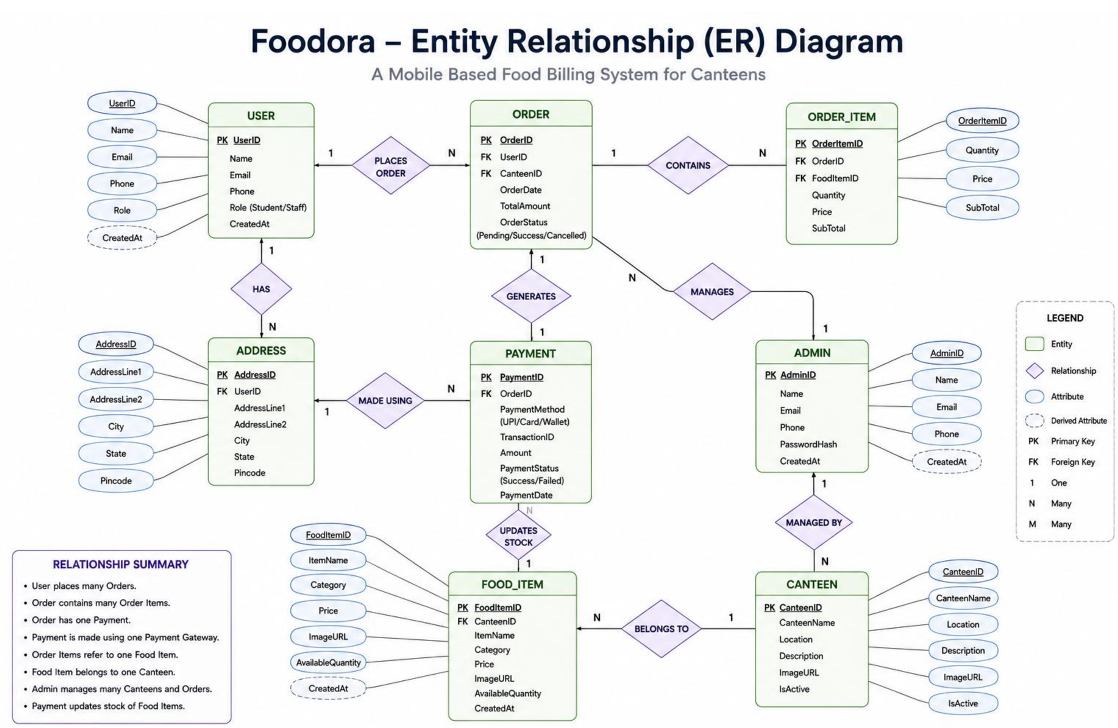


Figure 4.3 – Entity Relationship (ER) Diagram

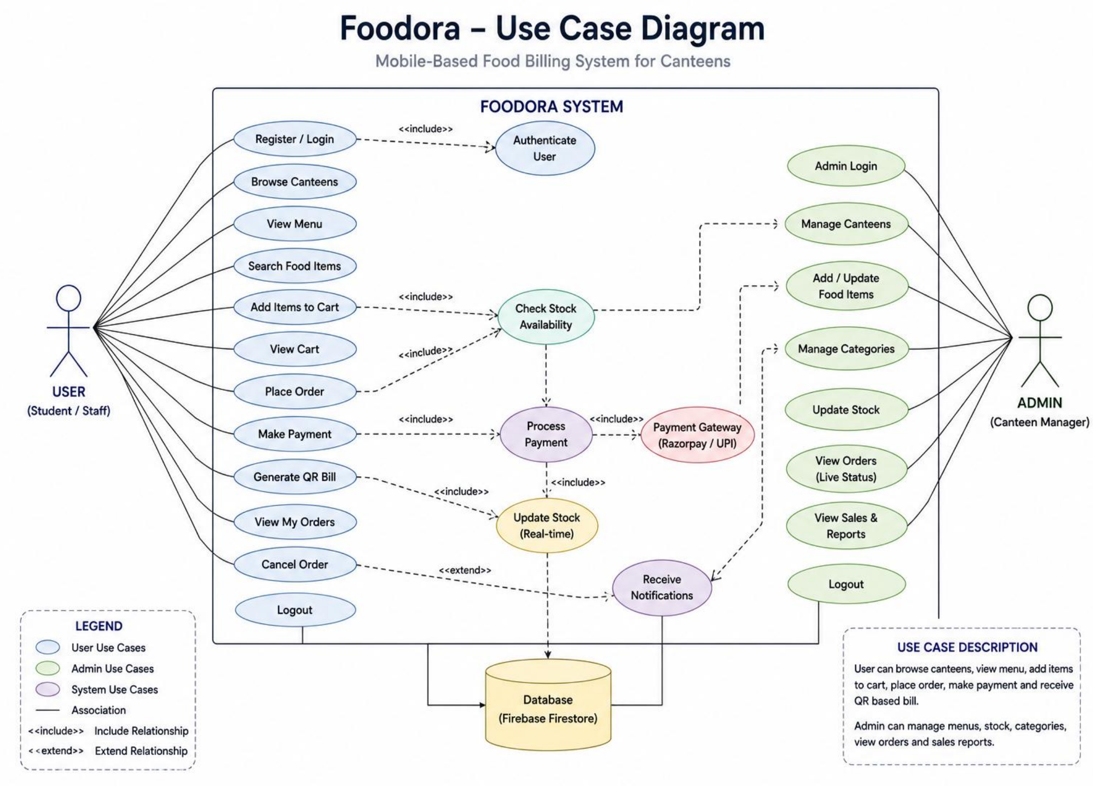


Figure 4.4 – Use Case Diagram

CHAPTER 5: MODULE DESCRIPTION

The Foodora application is divided into several functional modules to ensure organized system operation and efficient food management. Each module is designed independently while working together seamlessly within the application.

5.1 Home Module

The Home Module acts as the entry point of the application. It displays all available canteens and food places using a grid-based layout with two locations per row. Each canteen card contains:

- Canteen name
- Thumbnail image
- Availability status

This module allows users to quickly browse and select their preferred food location for ordering. The simple and intuitive interface improves navigation and enhances user experience.

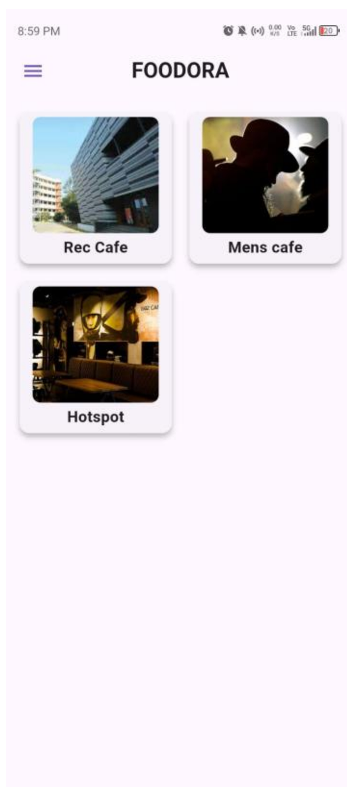


Figure 5.1 – Home Module – Canteen Listing Screen

5.2 Food Selection Module

The Food Selection Module enables users to browse and order food items easily.

Features

- Search bar for filtering food items
- Horizontally scrollable category chips
- Food item cards with image, name, price, and availability
- Quantity adjustment using + / – buttons
- Real-time total price calculation
- Checkout option

Food categories such as Snacks, Meals, and Beverages help users quickly find desired items. The checkout section dynamically updates the total bill amount based on selected items.

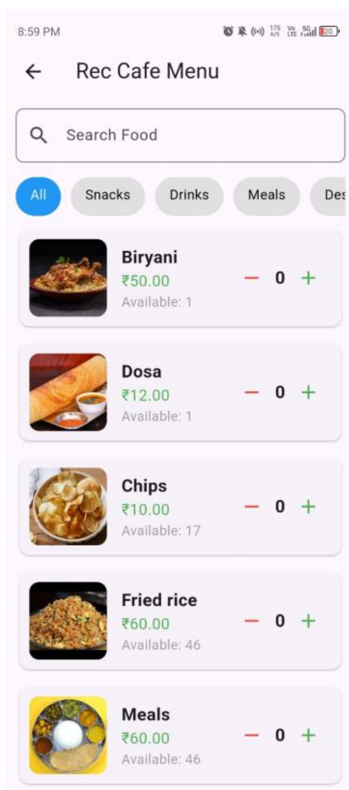


Figure 5.2 – Food Selection Module – Menu and Category Screen

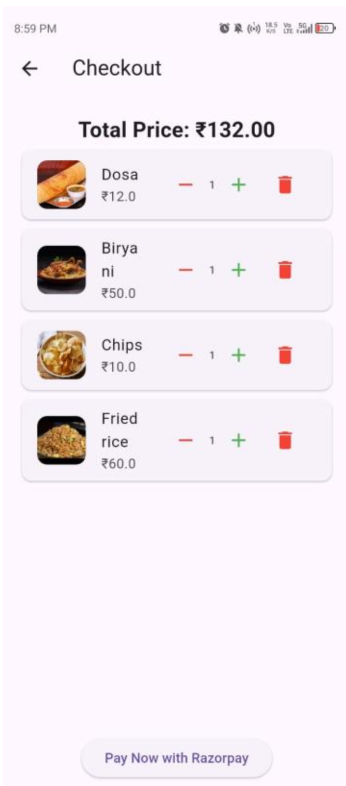
5.3 Checkout and Billing Module

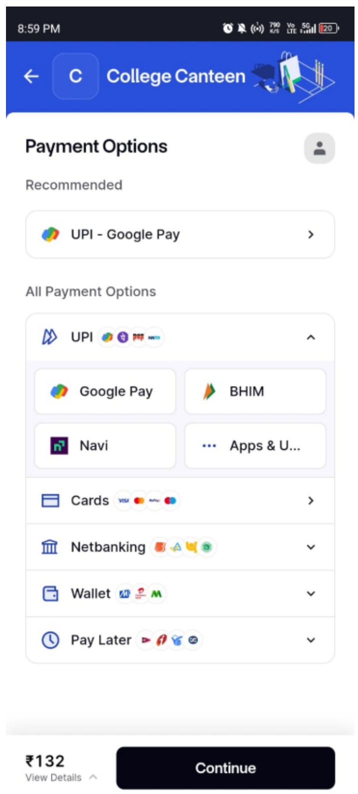
The Checkout and Billing Module handles payment processing and bill generation.

Features

- QR-code generation for order verification
- Razorpay and UPI payment integration
- Secure transaction handling
- Real-time stock updates after successful payment
- Automatic rollback if payment fails

After payment completion, a QR code is generated instantly and stored within the application for future access and order verification. This module ensures fast and secure digital transactions.





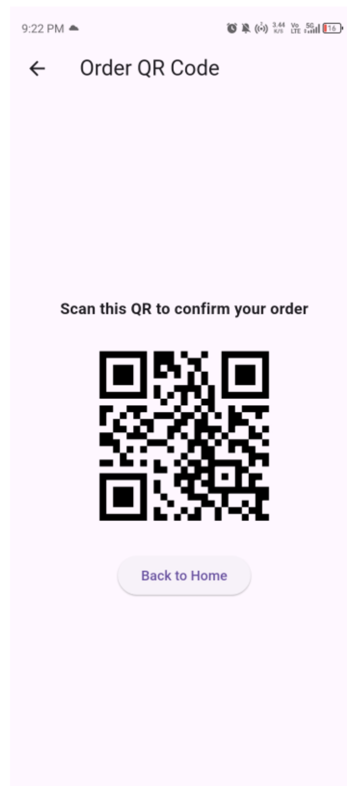


Figure 5.3 – Checkout Module – Order Summary Screen

Figure 5.4 – Checkout Module – Razorpay Payment Gateway Screen

Figure 5.5 – Checkout Module – QR Code Bill Generation Screen

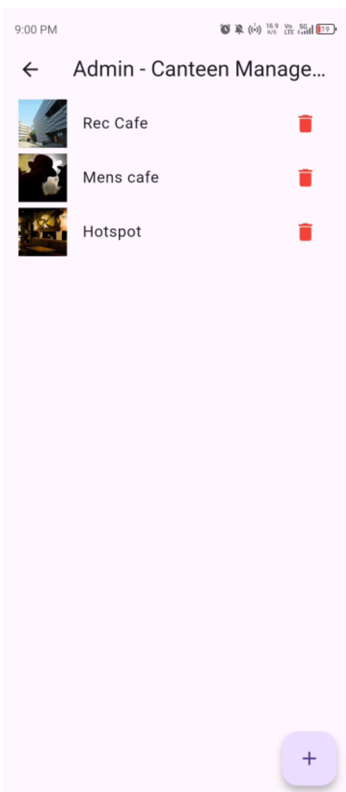
5.4 Admin Module

The Admin Module allows administrators to manage the overall food ordering system efficiently.

Features

- Add or update food places
- Add, edit, or remove food items
- Manage stock quantity and prices
- Monitor live order status
- View sales reports and stock trends

The module helps canteen staff maintain accurate inventory records and improve operational management.



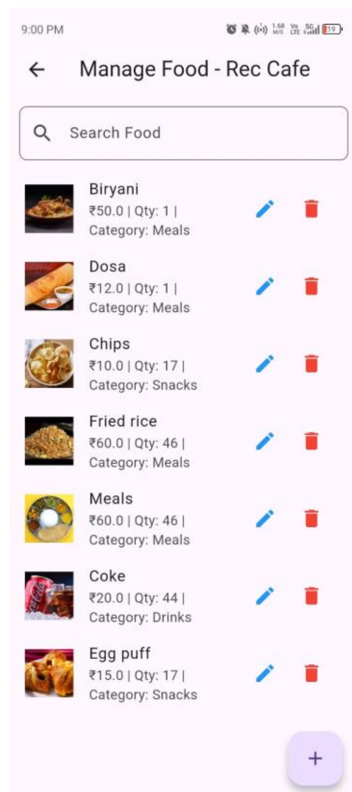


Figure 5.6 – Admin Module – Dashboard Screen

Figure 5.7 – Admin Module – Food Item Management Screen

5.5 My Orders Module

The My Orders Module stores all completed orders placed by the user.

Features

- Display order history
- View QR-code bills
- Access order details
- Delete or cancel orders

Whenever a payment is successful, the generated QR code and order details are automatically stored within this section for future reference.

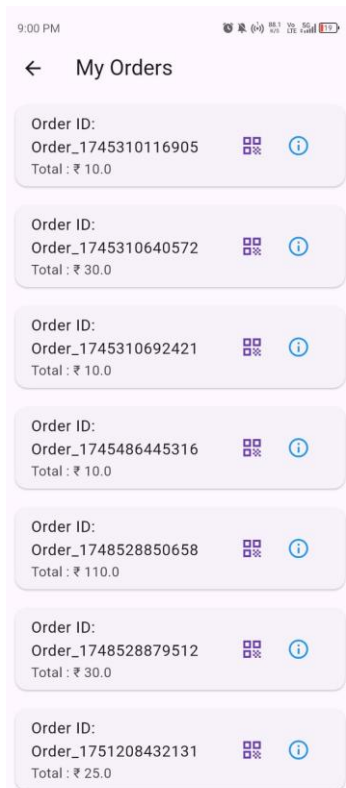


Figure 5.8 – My Orders Module – Order History Screen

5.6 Key Benefits of the Proposed System

Faster Ordering Process: The application significantly reduces food ordering time compared to traditional methods.

Real-Time Inventory Management: Stock quantities are updated automatically after every order, reducing inventory errors.

Secure Digital Payments: Razorpay and UPI integration ensure safe and reliable transactions.

Improved User Experience: The modern UI design and fast navigation provide a seamless ordering experience.

Reduced Manual Errors: Automation minimizes billing mistakes and inventory inconsistencies.

Contactless Ordering: QR-code-based billing supports contactless food collection and verification.

CHAPTER 6: EXPERIMENTAL TEST AND RESULTS

Experimental testing was conducted in a simulated college canteen environment to evaluate the performance, functionality, and reliability of the Foodora application. Testing was performed using Wi-Fi and 4G mobile networks to analyze application responsiveness under different conditions.

During testing, food search and filtering worked accurately, quantity adjustments updated instantly, QR-code generation completed successfully, payment transactions processed securely, and inventory updates reflected in real time.

6.1 Performance Metrics

Metric	Average Time	Maximum Time
App Response Time	300 ms	1.3 s
QR Code Generation	< 1 s	< 1 s
Payment Confirmation	2.5 s	2.5 s

Table 6.1 – Foodora Performance Metrics

The results demonstrate strong backend performance and minimal delay during concurrent user activity.

6.2 Comparison with Existing Systems

Operation	Foodora	Kiosk System	Manual Method
Order Placement	3.1 s	7.2 s	55 s
Payment Processing	2.5 s	6.2 s	30 s
QR Code Generation	0.8 s	1.5 s	N/A

Table 6.2 – Speed Comparison: Foodora vs Kiosk vs Manual

Foodora significantly outperformed kiosk and manual systems in speed and efficiency.

6.3 Success Rates

Parameter	Foodora	Kiosk System	Manual Method
Order Success Rate	95%	85%	75%
Payment Success Rate	98%	90%	85%

Table 6.3 – Success Rate Comparison

The application achieved high order and payment success rates, demonstrating system reliability and user satisfaction.

6.4 Error and Crash Rates

System Type	Error / Crash Rate
Foodora App	< 1%
Kiosk System	~2.5%
Manual Method	~8%

Table 6.4 – Error and Crash Rate Comparison

The low error rate indicates stable performance and readiness for real-world deployment.

CHAPTER 7: CONCLUSION AND FUTURE ENHANCEMENTS

Foodora provides a practical, scalable, and efficient solution for digitizing food ordering and billing systems in college campuses. The application integrates food ordering, QR-code billing, digital payments, and inventory management into a single mobile platform.

The system successfully reduces waiting time, improves stock management, minimizes manual errors, and enhances user satisfaction through automation and real-time synchronization.

7.1 Advantages of Foodora

- Faster ordering process
- Secure online payments
- Real-time stock updates
- QR-code-based billing
- Improved operational efficiency
- Better customer experience

7.2 Future Enhancements

The following enhancements are planned for future development iterations of Foodora:

- Loyalty reward system — points accumulation and redemption for frequent users
- Push notification support — order status alerts and canteen promotions
- AI-based food recommendations — personalized suggestions based on order history
- Real-time analytics dashboard — sales trends and inventory insights for administrators
- Multi-language support — regional language interfaces for broader accessibility
- Voice-based ordering system — hands-free ordering using voice commands
- Expansion to offices, hospitals, and event food courts

These future improvements can further enhance scalability and user engagement, positioning Foodora as a versatile digital food management platform beyond college campuses.

CHAPTER 8: REFERENCES

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